

CSE 232

Computer Networks

Homework-5

Group 15

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Assumptions

- The **LIST** command returns all client IDs, including the client who sent the command.
- The **DATA** command never includes invalid client IDs, and the message format is always correct.

Code

- The **recv_message** function always **printf** the received message to the server's terminal. If the message doesn't start with either of these prefixes, it does nothing.
- if (msg[0] == 'L' && msg[1] == 'I' && msg[2] == 'S' && msg[3] == 'T'), we iterate over **valid_ids** array and append indices (i.e. client IDs) whose entries are not "-1" to the string using **sprintf** and **strcat**. After this, the string is then sent to the requesting client using **send_message**.
- if (msg[0] == 'D' && msg[1] == 'A' && msg[2] == 'T' && msg[3] == 'A'), then it reads the number of recipient IDs into **num_rcv** from the command. Then, it dynamically allocates an array with **num_rcv** entries. It then reads all the recipient IDs into the array. Then, it extracts the message to be transmitted to recipients from the command. Finally, it iterates over the recipient IDs and sends them the transmission using **send_message**.

```
akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ make
gcc -Wall -Werror -O3 -o client client.c
gcc -Wall -Werror -O3 -o server server.c server_helper.c
akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./server
client 0 connected
client 1 connected
client 2 connected
client 3 connected
No activity for 30 secs.
client 4 connected

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.
```

Connection setup between the server and all 5 clients.

Q1. What is the output on all clients' and the server's terminals when the fifth client enters? LIST

```
akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./server
client 0 connected
client 1 connected
client 2 connected
client 3 connected
client 4: LIST

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
LIST
server: Available Clients : 0 1 2 3 4

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
LIST
server: Available Clients : 0 1 2 3 4

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs$ ./client
LIST
server: Available Clients : 0 1 2 3 4
```

- When **LIST** is executed in the 5th client, it gets all available clients from the server, including their numbers.
- Server gets output of “LIST” sent from the 5th client.
- No other client gets any output.

Q2. What is the output on all clients' and the server's terminals when the fifth client enters DATA 3 3 1 2: hello world

```

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs
akshat@akshat-IdeaPad-3-15IIL05-Ua:~/Downloads/CN-master/hw_cs$ ./server
★ client 0 connected
client 1 connected
client 2 connected
client 3 connected
client 4 connected
client 4: DATA 3 3 1 2: hello world

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs
akshat@akshat-IdeaPad-3-15IIL05-Ua:~/Downloads/CN-master/hw_cs$ ./client
No activity for 30 secs.

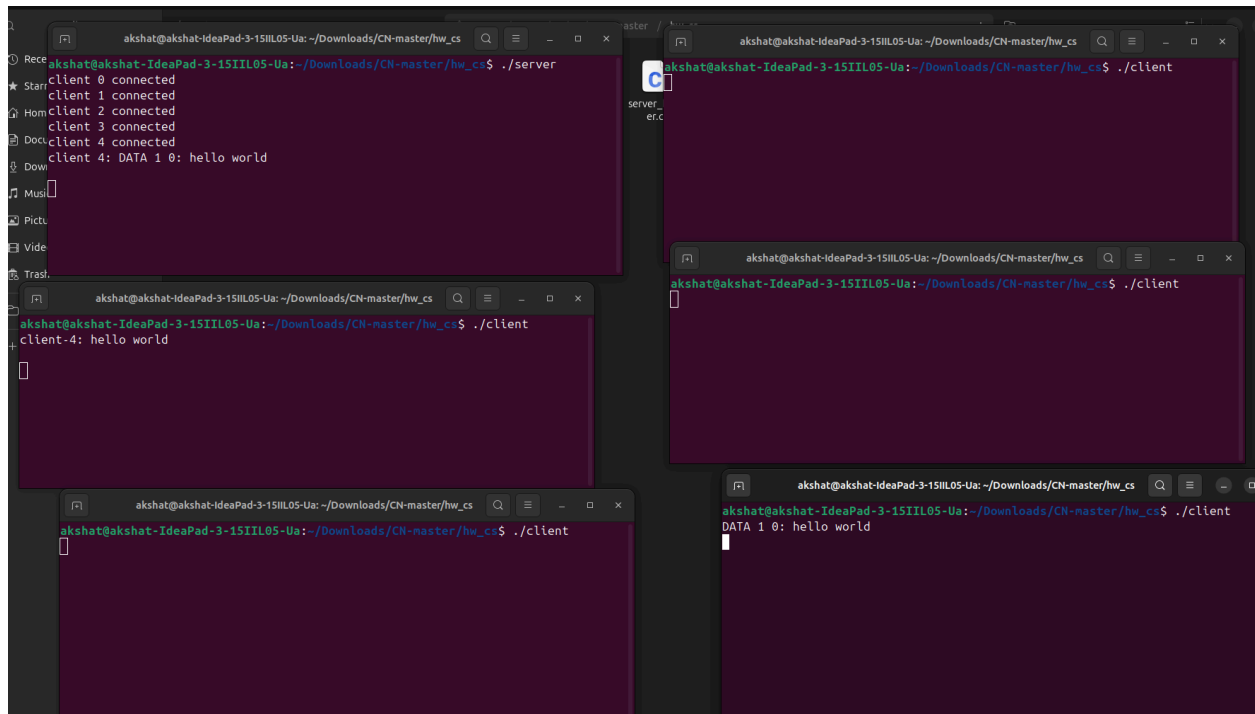
akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs
akshat@akshat-IdeaPad-3-15IIL05-Ua:~/Downloads/CN-master/hw_cs$ ./client
client-4: hello world

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs
akshat@akshat-IdeaPad-3-15IIL05-Ua:~/Downloads/CN-master/hw_cs$ ./client
DATA 3 3 1 2: hello world

```

- Server gets notified that the 5th client has sent the command “DATA 3 3 1 2: hello world”.
- Only the 2nd, 3rd, and 4th clients get an output of “hello world”.
- The 1st client does not get any output.

Q3.What is the output on all clients' and the server's terminals when the fifth client enters DATA 1 0: hello world



```
akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs $ ./server
client 0 connected
client 1 connected
client 2 connected
client 3 connected
client 4: DATA 1 0: hello world

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs $ ./client

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs $ ./client

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs $ ./client

akshat@akshat-IdeaPad-3-15IIL05-Ua: ~/Downloads/CN-master/hw_cs $ ./client
DATA 1 0: hello world
```

- Only the 1st client gets an output “hello world” as it is numbered 0.
- No other clients(2nd, 3rd, 4th, 5th) got any output.
- The server got notified that the 5th client has sent the command “DATA 1 0: hello world”.